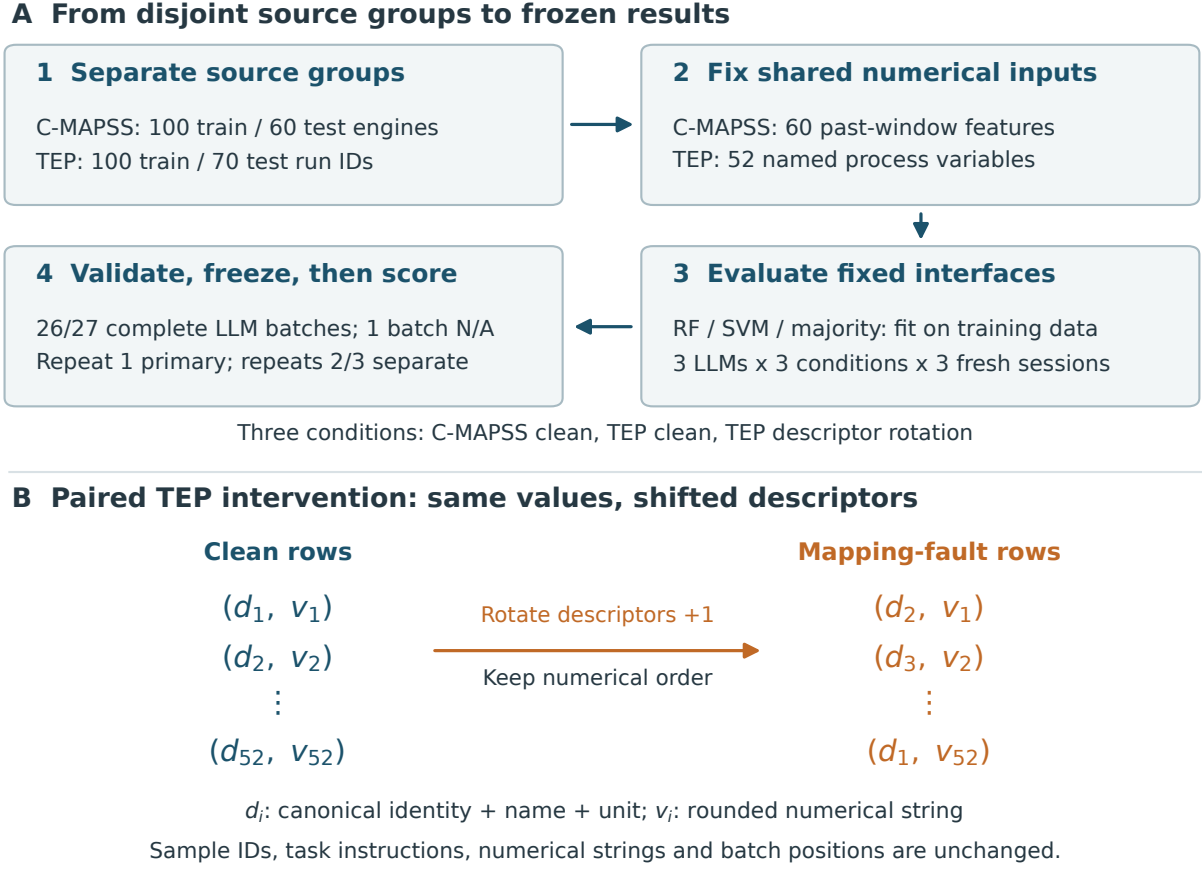
A From disjoint source groups to frozen results
1 Separate source groups
C-MAPSS: 100 train / 60 test engines
TEP: 100 train / 70 test run IDs
2 Fix shared numerical inputs
C-MAPSS: 60 past-window features
TEP: 52 named process variables
3 Evaluate fixed interfaces
RF / SVM / majority: fit on training data
3 LLMs x 3 conditions x 3 fresh sessions
4 Validate, freeze, then score
26/27 complete LLM batches; 1 batch N/A
Repeat 1 primary; repeats 2/3 separate
Three conditions: C-MAPSS clean, TEP clean, TEP descriptor rotation
B Paired TEP intervention: same values, shifted descriptors
Clean rows
(d1, v1)
(d2, v2)
(d52, v52)
Rotate descriptors +1
Keep numerical order
Mapping-fault rows
(d2, v1)
(d3, v2)
(d1, v52)
di: canonical identity + name + unit; vi: rounded numerical string
Sample IDs, task instructions, numerical strings and batch positions are unchanged.